Word union invariants and fast CRT certificates

Two producer components, 2026-11-01. Independent review pending; evidence counts remain separate.

Word union result. Literal Int8/UInt64 union-potential construction and reversal decode to the existing field-level formulas. Both orientations preserve valid edge equations and fixed-anchor equations after the parent, potential and optional parent-anchor update. Canonical coefficients are preserved by the modeled update. Fixed child anchors are transported using the literal signed subtraction formula.

Word union assumptions. Signs are plus or minus one, coefficients are canonical, the scalar modulus is prime and positive, and r is at most 2^63 where addition/subtraction is used. The field-state theorem assumes that the incoming reduced root equation is true. The current model covers arithmetic and graph/anchor writes; it does not include native size-vector updates, overflow of sizes/counters, returned constraint handling, or the proof that a distinct-root union preserves the terminating forest. Connecting the two preceding find calls to this update remains a composition task.

Word union checks. Six modules freshly replayed; eleven main declarations, four fixed controls and fifteen exact named axiom readbacks passed. A false theorem and three malformed-evidence checker controls were rejected. Fixed controls cover positive union sign, reversed offsets, negative anchor transport and invalid zero sign. Earlier failures involved explicit structure projections, a missing prime instance, an unavailable structure ext lemma, and unfolding Sign in a control. An early dependent check and first full replay also failed before those repairs; all receipts are retained. Final replay2 passed with unchanged bindings.

CRT result. Over an arbitrary commutative ring, the support certificate is (Ua-Ub)*H=C(norm)-C(aa squared)*Ub. The interpolation numerator certificate is (cc*X+dd)*H-(C(s0)+C(s1)*X)=Ub*C(-aa*cc). A third declaration substitutes every source coefficient difference literally and provides the corresponding divisibility witness. These identify the algebra behind the optimized branch's norm and slope formulas.

CRT checks. A separate agent compiled three main declarations and five fixed controls, then performed eight separate named axiom readbacks. The deliberately false theorem returned one. All successful receipts record unchanged bindings. The first main attempt omitted map_add normalization; its failed receipt remains. Controls include identical-support zero norm, nonzero norm with zero slope, repeated support with coprime comparison polynomial, ordinate orientation and source-parameter substitution. These are producer checks, not independent review or measured executions.

CRT scope. The identities themselves need only a commutative ring. Native arithmetic correspondence will require the prime-field path, absent diagnostic tables, canonical coefficients and its stated modulus bounds. Nonzero-norm coprimality, field interpolation, product divisibility, exact quotient, leading coefficient, batched inversion and full fast/fallback output equality are next obligations. No divisor-class addition result follows from these two identities alone.

Source and prior work. Native src/cryptanalysis/three_leads.rs at 55b56b1e04f2c4763fa1090d99961864dff221e8, SignedGraph edge union and optimized quadratic Jacobian addition. Existing GraphAlgebra, GraphUnion and word-find-composition supply the field and signed-word interfaces. The optimized-branch audit, archived alongside these packets, records the inspected source and primary literature. No novelty or performance claim is made.

Reproduction and next work. Each component retains sources, pinned source-host dependency paths, receipts and readbacks. Dependency caches are reused and omitted from publication; the archive is not a self-contained toolchain distribution. Next compose word union with find/forest/constraint results and derive the optimized branch interpolant from the CRT certificates. Frozen cohorts and original campaign limits are unchanged; no scientific run or new cost ratio is reported.